

Lecture 2: Summary Statistics

Mean Vector, Covariance and Correlation

STAT 481 — Multivariate Analysis

Thu, Aug 27, 2026 • 50 minutes

Today's Overview

- **Today's question:** two of the eight indicators in `worldhealth2022` are almost the same variable — which two?
- **Two symbols unlocked today:** **S**, the sample covariance matrix, and **R**, the sample correlation matrix.
- **The R:** `colMeans()`, `cov()`, `cor()`, `det()`, and our first picture of a matrix.
- **New data:** 175 countries, 8 indicators, World Bank 2022 — the data set the rest of the course is built on.
- **Next:** Sunday is the first R lab — bring your laptop.

Packages today

- **stats / base** [known] — `colMeans()`, `cov()`, `cor()`, `det()`: the mean vector, **S**, **R**, and the generalized variance. These *compute*.
- **No package for the data** [NEW] — `worldhealth2022.csv` is a plain text file in the course project, read with base R's `read.csv()`.
- **corrplot** [NEW] — one picture of a $p \times p$ correlation matrix. This one *shows*.

Install once with `install.packages("corrplot")`, or run `source("R/setup.R")` in the course project. **Base function for the concept, package for the picture:** `cor()` computes the matrix, `corrplot()` draws it. Open `STAT481.Rproj` first, so the relative path `"data/worldhealth2022.csv"` resolves.

Learning Objectives

By the end of this lecture, you should be able to:

- Compute $\bar{\mathbf{x}}$, **S** and **R** by hand for a small data array, in exact arithmetic.
- Use `colMeans()`, `cov()` and `cor()` in R, state the dimensions of each result before running it, and convert between **S** and the textbook's $\mathbf{S}_n$.
- Explain why **S** is scale-dependent and **R** is not, and interpret the generalized variance $|\mathbf{S}|$ — including what it hides.

1 Motivation: two of these are the same variable

Tuesday's answer, from the worksheet you were asked to bring — and why it was never about cities. You screened 41 US cities one column at a time: a screen that flags a city whenever any one of its 7 measurements is among the 3 most extreme caught **23 of 41**, i.e. 56.1%, against 67.0% if the seven columns had carried unrelated information. Observed *below* chance meant the flags were piling onto the same few cities — **the columns were not independent**. That is a statement about *columns*, not about Chicago and not about the United States. So today the data changes completely, and we watch the same thing happen, harder.

	Tuesday	Today, and the rest of the course
units	41 US cities, 1970	175 countries, 2022
variables	7 air-pollution measurements	8 health and development indicators
source	HSAUR2	<code>data/worldhealth2022.csv</code> (World Bank)

$\mathbf{X}$ is now 175×8 , so $n = 175$, $p = 8$, and 1400 numbers. Nobody can read 1400 numbers. The first thing anyone does with a data set is *summarise* it — and the interesting question is not how to summarise, but **what each summary throws away**.

Today's question

Among the eight indicators — `life_exp`, `infant_mort`, `fertility`, `pop65`, `urban_pct`, `gdp_pc`, `health_exp_pct`, `internet_pct` — **two are almost the same variable**. Which two, and how would you find out without being told?

Eight means will not answer it, and neither will eight variances: both describe the columns *one at a time*, while “almost the same variable” is a claim about a *pair*. That step — one-at-a-time to pairwise — is the step from univariate to multivariate statistics, and it is what today's two matrices take.

2 The mean vector

Definition 2.1: Sample mean vector

For the $n \times p$ data matrix $\mathbf{X}$ with rows $\mathbf{x}_1^\top, \dots, \mathbf{x}_n^\top$, the **sample mean vector** is the $p \times 1$ column vector of column averages,

$$\bar{\mathbf{x}} = \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i = (\bar{x}_1, \bar{x}_2, \dots, \bar{x}_p)^\top, \quad \bar{x}_k = \frac{1}{n} \sum_{i=1}^n x_{ik}.$$

It is the *centre of gravity* of the n points in $\mathbb{R}^p$: the single point that best represents where the cloud sits.

Two symbols above, both from Lecture 1: $\mathbf{x}_i^\top$ is row i of $\mathbf{X}$ stood up as a column, and $\mathbb{R}^p$ is the set of points with p coordinates — here one point, with 8 coordinates, per country.

If you missed Tuesday

Five rules from Lecture 1, all used below. Anything after a `#` is a comment and R ignores it. Quotation marks mark plain text — copy them exactly as printed. `W$life_exp` is one whole column of `W`. `W[i, j]` is row first, then column, and an empty slot means *all of them*. A leading `[1]` in output is R numbering the printed line, not part of the answer. Lecture 1, Section 4 works all five slowly.

```
wh <- read.csv("data/worldhealth2022.csv", row.names = 1)
W   <- wh[, 4:11]           # the eight numeric indicators; cols 1-3 are
                             labels
n   <- nrow(W); p <- ncol(W)
c(n = n, p = p)
```

```
  n    p
175    8
```

<code>life_exp</code>	life expectancy at birth, years	<code>urban_pct</code>	urban population, %
<code>infant_mort</code>	infant mortality, per 1000 births	<code>gdp_pc</code>	GDP per capita, US\$
<code>fertility</code>	births per woman	<code>health_exp_pct</code>	health spending, % of GDP
<code>pop65</code>	population aged 65+, %	<code>internet_pct</code>	Internet users, %

Six new things in four lines. `<-` stores a result under a name, so `wh` means that whole table from now on. `read.csv()` reads a plain text table off disk; `row.names = 1` says *use the first column as the row labels*, which is why the country codes do not become a ninth variable. `wh[, 4:11]` is Lecture 1's bracket with a *range* of columns in the second slot: columns 1 to 3 hold **country**, **region** and **income**, which are text, and a covariance needs numbers. `nrow()` and `ncol()` return the two halves of Lecture 1's `dim()` one at a time, and the semicolon simply puts two commands on one line. Finally `c()` glues loose values into a single sequence, and writing `c(n = n, p = p)` *names* them as it goes — which is where the `n` and `p` headings above the output came from.

```
round(colMeans(W), 2)
```

```
# the sample mean vector
```

life_exp	infant_mort	fertility	pop65	urban_pct
72.53	19.98	2.47	9.74	61.63
gdp_pc	health_exp_pct	internet_pct		
18450.92	6.82	69.56		

`round(x, 2)` is printing convenience: the second argument says how many decimals to *show*. R keeps every digit internally, so rounding never changes a stored number.

Eight numbers instead of 1400 — and already we have thrown almost everything away. The mean vector tells us *where* the cloud is and nothing at all about its *shape*: two very different data sets can share a mean vector exactly.

3 The covariance matrix, once by hand

Notation

S is the **sample covariance matrix**, **R** the **sample correlation matrix** — both $p \times p$ and symmetric, both built from the $n \times p$ table one *pair of columns* at a time:

$$s_{jk} = \frac{1}{n-1} \sum_{i=1}^n (x_{ij} - \bar{x}_j)(x_{ik} - \bar{x}_k), \quad r_{jk} = \frac{s_{jk}}{\sqrt{s_{jj} s_{kk}}}.$$

$j = k$ gives s_{jj} , the ordinary variance of column j : **S** carries the p variances on its diagonal, **R** carries ones. In R: `cov()` and `cor()`. Section 4 unlocks and defines both properly.

We compute $\bar{\mathbf{x}}$, **S** and **R** by hand exactly once in this course — do it carefully and you have earned the right to type `cov()` thereafter.

Example 3.1 (Four students, two variables). Four students report X_1 = hours of revision and X_2 = mark out of 20 on a quiz:

$$\mathbf{X} = \begin{pmatrix} 2 & 8 \\ 4 & 9 \\ 4 & 11 \\ 6 & 12 \end{pmatrix}, \quad n = 4, \quad p = 2.$$

Compute $\bar{\mathbf{x}}$, **S**, **R** and $|\mathbf{S}|$ in exact arithmetic.

Solution. **Step 1 — the mean vector.** Column averages: $\bar{x}_1 = \frac{2+4+4+6}{4} = 4$ and $\bar{x}_2 = \frac{8+9+11+12}{4} = 10$, so $\bar{\mathbf{x}} = (4, 10)^\top$.

Step 2 — centre the columns. Subtract each column's mean:

$$\text{column 1: } (-2, 0, 0, 2), \quad \text{column 2: } (-2, -1, 1, 2).$$

Always check: each centred column must sum to zero. $-2 + 0 + 0 + 2 = 0$; $-2 - 1 + 1 + 2 = 0$. Good.

Step 3 — the three distinct entries of S, dividing by $n - 1 = 3$:

$$s_{11} = \frac{(-2)^2 + 0^2 + 0^2 + 2^2}{3} = \frac{8}{3}, \quad s_{22} = \frac{(-2)^2 + (-1)^2 + 1^2 + 2^2}{3} = \frac{10}{3},$$

$$s_{12} = s_{21} = \frac{(-2)(-2) + (0)(-1) + (0)(1) + (2)(2)}{3} = \frac{4 + 0 + 0 + 4}{3} = \frac{8}{3}.$$

$$\text{So } \mathbf{S} = \frac{1}{3} \begin{pmatrix} 8 & 8 \\ 8 & 10 \end{pmatrix} = \begin{pmatrix} 2.\bar{6}\bar{6} & 2.\bar{6}\bar{6} \\ 2.\bar{6}\bar{6} & 3.\bar{3}\bar{3} \end{pmatrix}.$$

Step 4 — the correlation.

$$r_{12} = \frac{s_{12}}{\sqrt{s_{11} s_{22}}} = \frac{8/3}{\sqrt{(8/3)(10/3)}} = \frac{8}{\sqrt{80}} = \frac{2}{\sqrt{5}} = 0.894427\dots, \quad \mathbf{R} = \begin{pmatrix} 1 & 0.8944 \\ 0.8944 & 1 \end{pmatrix}.$$

Note that the $\frac{1}{3}$ cancelled: *the divisor never affects a correlation*.

Step 5 — the generalized variance. The bars mean *determinant*, not size; for a 2×2 matrix it is $ad - bc$, and Section 7 says what it is for. $|\mathbf{S}| = \frac{8}{3} \cdot \frac{10}{3} - \left(\frac{8}{3}\right)^2 = \frac{80 - 64}{9} = \frac{16}{9} = 1.777\bar{7}$.

Now the same thing in R — and every number must match:

```
X <- matrix(c(2, 8,
              4, 9,
              4, 11,
              6, 12), ncol = 2, byrow = TRUE) # byrow: fill row by row
colMeans(X)
cov(X) # divides by n - 1
cor(X)
```

```
[1] 4 10
      [,1] [,2]
[1,] 2.666667 2.666667
[2,] 2.666667 3.333333
      [,1] [,2]
[1,] 1.0000000 0.8944272
[2,] 0.8944272 1.0000000
```

Two new pieces of syntax there: `matrix()` folds a sequence into a table, and `ncol = 2, byrow = TRUE` are *named* arguments, so their order does not matter. In the output R labels a matrix rather than numbering it: `[,1]` heads column 1 and `[1,]` starts row 1 — the same row-then-column comma as `d[i, j]`.

$2.666667 = 8/3$, $3.333333 = 10/3$, $0.8944272 = 2/\sqrt{5}$: hand and machine agree to the last printed digit. That is the point of the exercise — from here on, when `cov()` prints something, you know what it did.

△! `byrow = TRUE`

`matrix()` fills **column by column** unless you tell it otherwise. Writing the data out in rows, as it appears on the page, and forgetting `byrow = TRUE` silently produces the *transpose* of what you meant — and if n and p happen to be compatible, R will not complain. It is the transpose trap of Lecture 1, and the reason you `dim()` a matrix straight after building it.

4 Unlocking S and R

★ NEW NOTATION UNLOCKED

S — the sample covariance matrix. **R** — the sample correlation matrix. The first two symbols this course unlocks — and the reason we built the $n \times p$ array on Tuesday at all.

Almost everything left in the course is *something you do to S or to R*:

- Lecture 5 (Sep 3) **inverts** it, to measure distance properly.
- Lecture 7 (Sep 15) **decomposes** it, to find the directions the data spreads in.
- Lecture 21 (Nov 15) **diagonalises** it — that is principal components.

Definition 4.1: Sample covariance and correlation matrices

$$\mathbf{S} = \frac{1}{n-1} \sum_{i=1}^n (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^\top, \quad s_{jk} = \frac{1}{n-1} \sum_{i=1}^n (x_{ij} - \bar{x}_j)(x_{ik} - \bar{x}_k),$$

$$\mathbf{R} = (r_{jk}), \quad r_{jk} = \frac{s_{jk}}{\sqrt{s_{jj} s_{kk}}}.$$

Use the second, entry-by-entry formula: that is the one you compute with. The first packs the same arithmetic into a single matrix product, which Lecture 6 unpacks.

Definition 4.1: Sample covariance and correlation matrices (continued)

Both are $p \times p$ and **symmetric**. The diagonal of $\mathbf{S}$ holds the p sample variances; the diagonal of $\mathbf{R}$ is all ones. $\mathbf{S}$ is also **positive semi-definite**, a property Lecture 7 explains and Lecture 5 needs.

⚠ The divisor: this course uses $n - 1$, the textbook uses n

Johnson & Wichern define the covariance with divisor n and write the result $\mathbf{S}_n$. R's `cov()` divides by $n - 1$. **This course always means the $n - 1$ version and always writes it $\mathbf{S}$.**

The two are related exactly, so you can always move between them:

$$\mathbf{S}_n = \frac{n-1}{n} \mathbf{S}, \quad \text{equivalently} \quad \mathbf{S} = \frac{n}{n-1} \mathbf{S}_n.$$

For the four students above, $\frac{n-1}{n} = \frac{3}{4}$ and $\mathbf{S}_n = \begin{pmatrix} 2 & 2 \\ 2 & 2.5 \end{pmatrix}$; Appendix A.1 confirms it in R.

Say which one you mean, every time. A number that disagrees with the textbook by a factor $n/(n-1)$ is almost always this and not a mistake. And **$\mathbf{R}$ is the same either way** — the divisor cancels, as Step 4 showed.

5 $\mathbf{S}$ on the real data — and why it will not answer our question

`diag()` pulls the diagonal out of a square matrix, so the line below gives the p sample variances — R working inside out: `cov()`, then `diag()`, then `round()`.

Now you — Checkpoint 1

Before I run it: which of the eight variables will have the largest variance, and roughly how many times larger than the smallest — 10? 1000? more? Write one number down, then look.

```
round(diag(cov(W)), 1)           # the eight sample variances
```

life_exp	infant_mort	fertility	pop65	urban_pct
67.3	517.7	1.6	51.0	487.4
gdp_pc	health_exp_pct	internet_pct		
801635374.1	10.7	648.6		

The problem announces itself, and louder than on Tuesday's data: `gdp_pc` has variance 801 635 374 and `fertility` has 1.6 — a ratio of half a *billion* to one. That says nothing about which variable matters, only that one is counted in dollars and the other in children per woman. Push it one step further: `gdp_pc` alone is 100.00% of $\text{tr}(\mathbf{S})$ to two decimals — $\text{tr}(\mathbf{S})$, the *total variance*, being the sum of those eight numbers, which Section 7 defines properly — so the other seven indicators together are a rounding error in it. **A covariance analysis of all eight columns would be an analysis of `gdp_pc` and nothing else.**

The same is true off the diagonal, where the entries are covariances rather than variances. The covariance between `infant_mort` and `gdp_pc` is $-252\,925.7$, and asking whether that is a *strong* relationship is not an answerable question: it is measured in (deaths per 1000 births) $\times$ (US dollars). Quote GDP per capita in thousands of dollars instead and the same covariance becomes -252.9 , with the relationship completely unchanged. Appendix A.2 prints the 3×3 block these two numbers are read from.

Fact 5.1: Covariance is scale-dependent; correlation is not

s_{jk} carries the units of variable j times the units of variable k and is unbounded, so **covariances cannot be compared across different pairs of variables** — which is precisely what our question requires.

Fact 5.1: Covariance is scale-dependent; correlation is not (continued)

Dividing by both standard deviations removes the units and bounds the result: $-1 \leq r_{jk} \leq 1$ always. In matrix form, with $\mathbf{D} = \text{diag}(\sqrt{s_{11}}, \dots, \sqrt{s_{pp}})$,

$$\mathbf{R} = \mathbf{D}^{-1} \mathbf{S} \mathbf{D}^{-1},$$

which is exactly the statement “ $\mathbf{R}$ is the covariance matrix of the standardised data”. It needs no matrix algebra: $\mathbf{D}$ is diagonal, so the formula says only *divide entry (j, k) by $\sqrt{s_{jj}}$ and $\sqrt{s_{kk}}$* (Lecture 6 does inverses properly). $\mathbf{R}$ agrees in one line, `cov2cor()` doing exactly that division and `all.equal()` asking whether two objects match to within floating-point tolerance — so `TRUE` means all 64 entries agree, not just one:

```
> all.equal(cov(W), cov2cor(cov(W)))
[1] TRUE
```

That `>` is R’s prompt: the line beside it is what you type, the line below is what comes back.

⚠ Never read a correlation off the covariance matrix

s_{jk} and r_{jk} sit in the same position of two matrices that print identically, and only r_{jk} is bounded by ± 1 . If you report “a correlation of -252.926 ”, you have read the wrong matrix. Worse, if you feed `cov()` output to a function that expects correlations — including `corrplot()` — it will run without error and draw a picture that is simply wrong. **Check the diagonal: a correlation matrix has ones on it.**

6 R, and the answer to today’s question

```
options(width = 100)      # 8 columns do not fit R's default width of 80
round(cov(W), 2)
```

	life_exp	infant_mort	fertility	pop65	urban_pct	gdp_pc	health_exp_pct	internet_pct
life_exp	1.00	-0.92	-0.81	0.66	0.55	0.56	0.17	0.77
infant_mort	-0.92	1.00	0.78	-0.56	-0.46	-0.39	-0.15	-0.74
fertility	-0.81	0.78	1.00	-0.67	-0.50	-0.42	-0.21	-0.84
pop65	0.66	-0.56	-0.67	1.00	0.43	0.59	0.36	0.61
urban_pct	0.55	-0.46	-0.50	0.43	1.00	0.52	0.23	0.72
gdp_pc	0.56	-0.39	-0.42	0.59	0.52	1.00	0.16	0.52
health_exp_pct	0.17	-0.15	-0.21	0.36	0.23	0.16	1.00	0.20
internet_pct	0.77	-0.74	-0.84	0.61	0.72	0.52	0.20	1.00

Without that first line R would break the matrix into two blocks — columns 1–6, then the remaining two — because it wraps output at 80 characters by default. `options(width = 100)` is a session setting, not a property of the data.

There it is: $r_{\text{life_exp}, \text{infant_mort}} = -0.92$. **Life expectancy and infant mortality are almost the same variable** — and the minus sign does not weaken that, it only says they run in opposite directions. Obvious in hindsight: the places where babies die are the places where people die young.

⚠ What a correlation of -0.92 actually costs you

It means these two columns carry *nearly the same information*, so any analysis that treats them as two independent pieces of evidence is double-counting. If your conclusion is “Internet access goes with long life”, an equally good fit to the data is “Internet access goes with low infant mortality” — the data cannot separate them. That is **confounding**, detected with one function call in the second week of the semester.

Note also what a *negative* near-duplicate does to the word “same”. Nothing in the arithmetic of $\mathbf{R}$ cares about the sign: $r = -0.92$ and $r = +0.92$ are equally redundant, because either one lets you predict the column you dropped.

△! What a correlation of -0.92 actually costs you (continued)

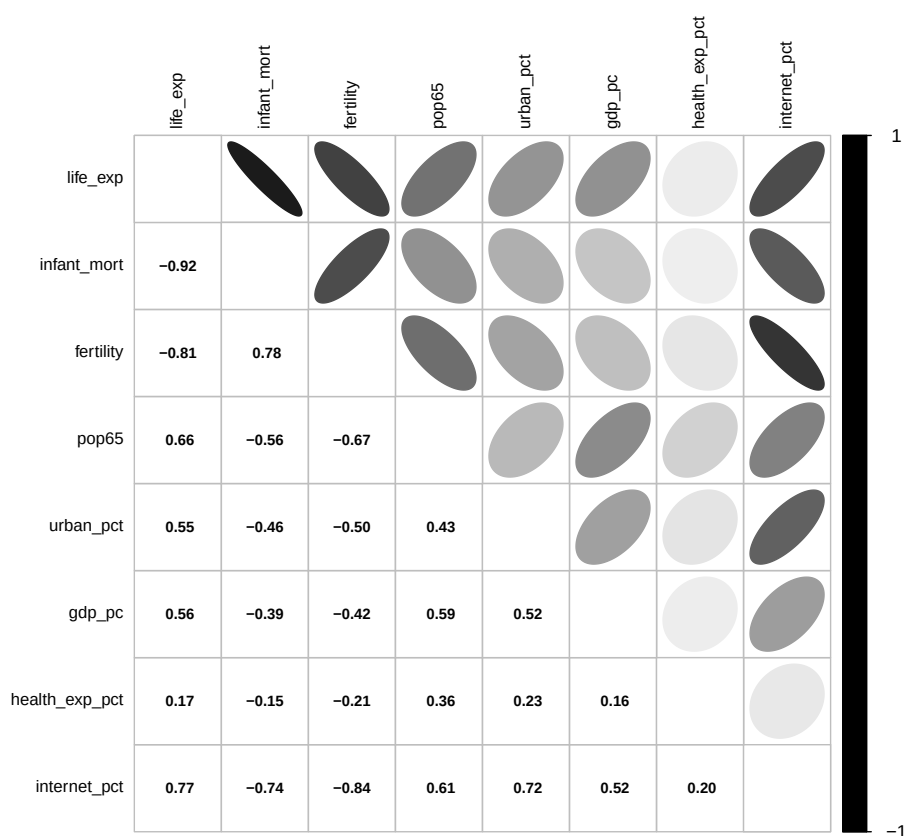
And the warning in the other direction. r measures *linear* association only. An r near zero says the cloud has no straight-line tilt; it does **not** say the two variables are unrelated. Lecture 4 (Tue, Sep 1) shows you a pair whose correlation rises from 0.56 to 0.79 the moment you take logarithms — the association was there the whole time; only the straight line was missing.

A matrix you can look at

Reading an 8×8 table is work; reading a picture is not. This is the first time in the course that a *matrix* becomes something you look at.

`corrplot.mixed()` is one function whose name happens to contain a dot; `upper =` and `lower =` are named arguments choosing how each triangle is drawn.

```
library(corrplot)
corrplot.mixed(cor(W), upper = "ellipse", lower = "number")
```



The 8×8 correlation matrix of *worldhealth2022*, drawn: the same numbers as the table above, arranged the same way. Full code in *L2_STAT481.qmd*.

How to read this figure

- **One cell is one correlation** — the same r_{jk} printed in the table above, in the same row and column.
- **Orientation carries the sign.** An ellipse slanting up is a positive r ; slanting down is negative.
- **Narrowness carries the strength.** The closer $|r|$ is to 1, the thinner the ellipse; a circle is r near 0.
- **The bar on the right is the legend**, running from -1 at the bottom to $+1$ at the top. It is the scale the shading is read against — not part of the matrix.
- **The diagonal is blank on purpose.** $r_{jj} = 1$ for every variable, so those eight cells carry no information at all.

How to read this figure (continued)

- **The printed version is grey** so that nothing depends on colour. On your screen the same command comes out in colour, and that is the correct result, not a mistake.

Look at the `life_exp-infant_mort` cell: it has collapsed to a line, tilted *downwards* because the correlation is negative. You found it before you read a single number, which is the whole argument for the picture. One more reading, in class: `internet_pct` is narrow against every column except `health_exp_pct` — an indicator of development behaves like a summary of all the others, and the one variable it fails to summarise is the same one that supplies the weakest correlation in the whole matrix. Two further pairs are visible at a glance, and Home Exercise 3(b) asks you to name them.

7 One number for the whole spread: the generalized variance

$\mathbf{S}$ has $p(p+1)/2 = 36$ distinct entries for our eight variables — eight variances plus 28 pairs. Is there a single number that says how spread out the data is overall?

Definition 7.1: Generalized variance and total variance

The **generalized variance** of a data set is the determinant of its covariance matrix, $|\mathbf{S}|$. The **total variance** is the trace, $\text{tr}(\mathbf{S}) = s_{11} + \cdots + s_{pp}$.

For an 8×8 , `det()` does the work. For the four students of Section 3, $|\mathbf{S}| = \frac{16}{9} = 1.78$.

```
det(cov(W))      # generalized variance of the raw data
det(cor(W))      # the same thing for the standardised data
```

```
[1] 8.935722e+18
[1] 0.001167422
```

`8.935722e+18` is R's shorthand for 8.935722×10^{18} ; an `e-04` would mean $\times 10^{-4}$.

$|\mathbf{S}| = 8.94 \times 10^{18}$ is **uninterpretable on its own** — it is built from eight variances in eight different squared units, so there is no scale to judge it against. Appendix A.4 says what it is good for and what else it hides.

$|\mathbf{R}| = 0.00117$ is **very interpretable**, because a correlation matrix has no units. Its determinant is bounded: $|\mathbf{R}| = 1$ exactly when the variables are mutually uncorrelated, and $|\mathbf{R}| = 0$ when one variable is an exact linear combination of the others — a weighted sum of them that reproduces it exactly. Our value is a long way down towards zero.

Fact 7.1: The first quantitative hint that this course has a second half

$|\mathbf{R}| = 0.0012$ says these eight indicators are strongly **redundant**: a smaller set of variables would carry nearly all of what they say.

Finding that smaller set is **principal component analysis**, Lecture 21 (Sun, Nov 15) — and this data set told you it was coming in the second week of term.

Self-check (1 minute)

Without computing anything:

- `cov(W)` and `cor(W)` — what are their dimensions, and what is on each diagonal?
- I re-express `gdp_pc` in *thousands* of dollars. Which of $\bar{\mathbf{x}}$, $\mathbf{S}$, $\mathbf{R}$, $|\mathbf{S}|$ change?
- A classmate reports “the correlation between `life_exp` and `gdp_pc` is 1.30×10^5 ”. What did they do wrong?

(Answers in the summary below; cover them and try first.)

Wrap-up

Summary of Lecture 2

- $\bar{\mathbf{x}}$ says *where* the cloud is; $\mathbf{S}$ how it is *spread*; $\mathbf{R}$ which variables *move together*, free of units; $|\mathbf{S}|$ *how much overall* — while hiding the shape.
- **This course divides by $n - 1$ and writes $\mathbf{S}$.** The textbook divides by n and writes $\mathbf{S}_n = \frac{n-1}{n}\mathbf{S}$. $\mathbf{R}$ is the same either way.
- **Covariance is scale-dependent, correlation is not** — $\mathbf{R} = \mathbf{D}^{-1}\mathbf{S}\mathbf{D}^{-1}$ — and correlation is invariant only under *increasing* linear transformations.
- **The answer to today's question:** `life_exp` and `infant_mort`, $r = -0.92$. The sign is irrelevant to the redundancy — and it is the first sign that eight indicators are really about fewer things.
- **The R to remember:** `colMeans()`, `cov()`, `cor()`, `det()` — and `corrplot()` to look at what `cor()` computed.

Self-check answers: (a) both 8×8 ; `cov` has the eight variances on its diagonal, `cor` has eight ones. (b) $\bar{\mathbf{x}}$, $\mathbf{S}$ and $|\mathbf{S}|$ all change; $\mathbf{R}$ does not. (c) They read the covariance matrix instead of the correlation matrix — no correlation can exceed 1.

Next lecture (Sun, Aug 30): *Lecture 3 — the first R lab.* R, RStudio and Quarto from zero, ending in your first reproducible report. One thing to take with you: `cov()` accepted `W` without a murmur, but the matrix multiplication we will need later will refuse it. Sunday you find out why, and what `wh[, 4:11]` was quietly protecting you from. **Bring your laptop.**

Announcements:

- **End of Add/Drop is today, Thu Aug 27, 2026.**
- **Sunday Aug 30 is a lab.** R, RStudio and Quarto must already be installed, and `STAT481.Rproj` must open. See me today if not.
- **R Lab 1 goes out Sun, Aug 30** and is due **Sun, Sep 6.**
- **Quiz 1: Thu, Sep 10**, covering Lectures 1–5.

Appendix A — Read at home

Required reading, not optional. Nothing below needs anything that is not already in Sections 1–7, Home Exercise 3 is set on A.3 and A.4, and all four parts are in the scope of Quiz 1. They are here rather than in the body only because fifty minutes is fifty minutes.

A.1 The two divisors, confirmed in R

Section 4 stated the exact relation $\mathbf{S}_n = \frac{n-1}{n}\mathbf{S}$. For the four students of Section 3, $\frac{n-1}{n} = \frac{3}{4}$, and multiplying a matrix by a number multiplies every entry by it:

```
> (3/4) * cov(X)
      [,1] [,2]
[1,]    2  2.0
[2,]    2  2.5
```

— which is exactly the $\mathbf{S}_n = \begin{pmatrix} 2 & 2 \\ 2 & 2.5 \end{pmatrix}$ that Section 4 predicted. **A number that disagrees with the textbook by a factor $n/(n-1)$ is almost always this, and not a mistake.**

A.2 Covariance off the diagonal, and the units it carries

Section 5 quoted two numbers from the covariance matrix without printing it. Here is the 3×3 block they come from. The bracket keeps every row — the slot before the comma is empty — and three named columns, glued into one sequence by `c()` again:

```
round(cov(W[, c("life_exp", "infant_mort", "gdp_pc")]), 1)
```

	life_exp	infant_mort	gdp_pc
life_exp	67.3	-171.0	130238.0
infant_mort	-171.0	517.7	-252925.7
gdp_pc	130238.0	-252925.7	801635374.1

Three things to read off it. The **diagonal** repeats three of the eight variances of Section 5 — 67.3, 517.7, 801635374.1. The matrix is **symmetric** about that diagonal, as every covariance matrix is. And not one of the three off-diagonal numbers can be compared with another: -171.0 is in (years) $\times$ (deaths per 1000 births), $130\,238.0$ is in (years) $\times$ (US dollars), and $-252\,925.7$ is in (deaths per 1000 births) $\times$ (US dollars). Three numbers, three different units, no ranking possible. **That is the entire case for R.**

A.3 What “scale-free” does and does not mean

R is unchanged if you change a variable’s units. But “unchanged” has a condition attached that the textbooks often drop.

```
cor(W$gdp_pc, W$internet_pct)
cor(W$gdp_pc / 1000, W$internet_pct) # a = 1/1000, positive
cor(-W$gdp_pc, W$internet_pct) # a = -1, negative
```

```
[1] 0.5249379
[1] 0.5249379
[1] -0.5249379
```

Notice what one line did: `W$gdp_pc / 1000` divides all 175 values at once, and `-W$gdp_pc` negates all 175. R works on whole columns.

Fact 7.2: The invariance property, stated correctly

For a linear transformation $x \mapsto ax + b$, the sample correlation is **unchanged when $a > 0$ and changes sign when $a < 0$** .

Dollars to thousands of dollars has $a = 1/1000 > 0$ and the correlation with Internet use is 0.5249379 before and after, to every printed digit; negating GDP has $a = -1 < 0$ and it flips to -0.5249379 . Home Exercise 3(a) asks you why. A real $a < 0$ you will meet in the worksheet: rural % = 100 – urban %.

Many sources drop the sign condition. It is false without it.

A.4 |S| — what it is good for, and what it hides

Section 7 called $|S| = 8.94 \times 10^{18}$ uninterpretable on its own, and left it there. Two reasons, and one use.

The units. $|S|$ is built from eight variances measured in eight different squared units, so the number it produces is in no unit anyone can name. There is nothing to judge 8.94×10^{18} against.

The use, therefore, is comparative — is *this* group of countries more spread out than *that* one, measured the same way on the same variables? — and never absolute. That is a real use, and it is the only one.

△! |S| hides the shape

A determinant is one number summarising $p(p+1)/2$ of them, so it must lose information — and what it loses is the *shape*. Two data sets with completely different covariance structures can have the same |S| — Home Exercise 3(c) asks you to build such a pair. Never report a generalized variance as if it described the data.

Try at home (answers below; full solutions in the practice session)

Home Exercise 1. For the 4×3 data array $\mathbf{X} = \begin{pmatrix} 3 & 5 & 1 \\ 5 & 9 & 2 \\ 7 & 9 & 2 \\ 9 & 13 & 3 \end{pmatrix}$:

- (a) Compute $\bar{\mathbf{x}}$ and the three diagonal entries of $\mathbf{S}$ by hand.
- (b) Compute s_{12} and hence r_{12} , exactly.
- (c) Without computing it: what is the textbook's s_{11} under the $\mathbf{S}_n$ convention?

Home Exercise 2. Run this. In the course project:

- (a) Compute $\mathbf{R}$ for $\mathbf{W}$ and report the *weakest* correlation in the matrix (the one closest to zero, off the diagonal), with the pair of variables it belongs to. Read it off the matrix printed in Section 6.
- (b) Standardise the data with `scale()` (it centres each column, then divides by that column's standard deviation) and compute `cov(scale(W))`. Compare it with `cor(W)`. What do you find, and why?
- (c) Compute `tr(S)` with `sum(diag(cov(W)))` (inside out: `cov()`, then `diag()`, then `sum()`). Which single variable accounts for essentially all of it, and what does that tell you about analysing this data set on the covariance matrix?

Home Exercise 3. Read at home, then answer. Appendix A.3 and Appendix A.4.

- (a) Run Appendix A.3's three `cor()` lines. Why does dividing by 1000 leave the correlation alone while negating GDP does not?
- (b) Name the two further associations flagged in Section 6 — one involving `fertility`, one involving `pop65` — and say what each means.
- (c) Give two 2×2 covariance matrices with the same determinant but different shapes. Which cloud is long and thin?

Answers: 1) (a) $\bar{\mathbf{x}} = (6, 9, 2)^\top$; centred columns $(-3, -1, 1, 3)$, $(-4, 0, 0, 4)$, $(-1, 0, 0, 1)$, so $s_{11} = \frac{9+1+1+9}{3} = \frac{20}{3}$, $s_{22} = \frac{32}{3}$, $s_{33} = \frac{2}{3}$. (b) $s_{12} = \frac{12+0+0+12}{3} = 8$, so $r_{12} = 8/\sqrt{(20/3)(32/3)} = 8/(\frac{8\sqrt{10}}{3}) = \frac{3}{\sqrt{10}} = 0.9487$. (c) $\mathbf{S}_n$ divides by $n = 4$ rather than 3, so $s_{11} = \frac{20}{4} = 5$ — that is $\frac{3}{4} \times \frac{20}{3}$, the $\frac{n-1}{n}$ factor. 2) (a) `health_exp_pct` and `infant_mort`, $r = -0.1483$ (printed as -0.15): essentially no linear association — what a country spends on health, as a share of GDP, says almost nothing about whether its babies survive. (b) They are equal (up to floating-point error): standardising makes every variance 1, so the covariance matrix of the standardised data is $\mathbf{R}$. (c) `tr(S)` = 801 637 158, of which `gdp_pc` contributes 801 635 374 — 100.00% to two decimals, purely because of its units. Any method that works on $\mathbf{S}$ rather than $\mathbf{R}$ is therefore an analysis of `gdp_pc` alone. 3) (a) $a = 1/1000 > 0$ preserves it; $a = -1 < 0$ flips the sign. (b) `fertility` with `internet_pct` (-0.84), the countries where women have most children are the least connected; `pop65` with `life_exp` (0.66), populations that live longer are older populations. (c) `diag(4, 1)` and `diag(2, 2)` both have determinant 4; the first is long and thin, the second round.